

Table 1: Inferred hanging mass from the measured period, with $M = 209.2\text{ g}$ and $r = 12.00\text{ cm}$ fixed

Mass m (g)	Period T (s)	Inferred mass m_{obs} (g)	Error
19.85	2.2460 ± 0.0045	20.047 ± 0.080	0.99%
38.84	1.6150 ± 0.0037	38.77 ± 0.18	-0.17%
58.70	1.3180 ± 0.0036	58.22 ± 0.31	-0.82%
68.52	1.2120 ± 0.0036	68.84 ± 0.40	0.47%
78.04	1.1370 ± 0.0033	78.23 ± 0.45	0.24%

Table 2: Linear regression of T^{-2} against m in Experiment 1, and the resulting estimate of M

Quantity	Value
Slope s ($\text{kg}^{-1}\text{ s}^{-2}$)	9.827 ± 0.046
M_{obs} (g)	210.50 ± 0.98
M_{ref} (g)	209.2
Error	0.62%

Table 3: Measured period at different rotation radii, with m fixed at an unknown value and $M = 209.2\text{ g}$

Rotation radius r (cm)	Period T (s)
10.00	1.2280 ± 0.0039
11.00	1.2990 ± 0.0034
12.00	1.3490 ± 0.0038
13.00	1.4040 ± 0.0034
14.00	1.4580 ± 0.0041

Table 4: Linear regression of T^2 against r in Experiment 2, and the resulting estimate of m

Quantity	Value
Slope s ($\text{s}^2 \text{m}^{-1}$)	15.18 ± 0.33
m_{obs} (g)	55.5 ± 1.2
m_{ref} (g)	55.34
Error	0.33%

Table 5: Mass predicted from Experiment 1's fit line, using Experiment 2's period at $r = 12.00$ cm

Quantity	Value
m_{obs} (g)	55.62 ± 0.35
m_{ref} (g)	55.34
Error	0.50%

Table 6: Raw period measurements for each hanging mass in Experiment 1

Mass m (g)	Raw periods T (s)
19.85	2.23, 2.24, 2.23, 2.26, 2.26, 2.24, 2.25, 2.25, 2.25, 2.25
38.84	1.62, 1.61, 1.62, 1.62, 1.60, 1.62, 1.62, 1.61, 1.61, 1.62
58.70	1.32, 1.31, 1.31, 1.32, 1.32, 1.32, 1.32, 1.33, 1.31, 1.32
68.52	1.21, 1.22, 1.21, 1.21, 1.22, 1.21, 1.20, 1.21, 1.22, 1.21
78.04	1.14, 1.13, 1.14, 1.14, 1.13, 1.14, 1.14, 1.14, 1.14, 1.13

Table 7: Raw period measurements for each rotation radius in Experiment 2

Rotation radius r (cm)	Raw periods T (s)
10.00	1.23, 1.23, 1.24, 1.23, 1.22, 1.23, 1.23, 1.23, 1.21, 1.23
11.00	1.31, 1.30, 1.30, 1.30, 1.30, 1.30, 1.30, 1.29, 1.29, 1.30
12.00	1.35, 1.35, 1.36, 1.34, 1.36, 1.34, 1.35, 1.34, 1.35, 1.35
13.00	1.41, 1.41, 1.40, 1.40, 1.40, 1.40, 1.41, 1.40, 1.40, 1.41
14.00	1.47, 1.46, 1.46, 1.44, 1.45, 1.46, 1.46, 1.46, 1.47, 1.45